Spatio-Temporal Analysis of Water Temperatures in Long Island Sound: Comparing ARIMA and STARIMA Models

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1. Introduction & Data Description

Climate change has been rapidly warming the Earth's oceans for the past few decades (Cheng et al., 2019). Increasing sea temperatures are known to affect marine ecosystems and these changes may alter the behavior of some organisms (Doney et al., 2012). Day-to-day, changing water temperatures can affect fishing and major economic activity, impacting coastal communities (NOAA Fisheries, 2026). Given the importance of water temperature on small and large time scales, this study investigates the spatio-temporal dynamics of water temperatures in Long Island Sound and surrounding waters from January 2024 through December 2025. The Sound is a large ocean inlet, connected to the North Atlantic Ocean, tucked between Connecticut and New York State along the northeast coast of the United States.

High-frequency (6-minute interval) temperature data collected by six stations located in and near Long Island Sound were sourced from the United States' National Oceanic and Atmospheric Administration (NOAA). An API from NOAA's Center for Operational Oceanographic Products and Services (CO-OPS) was utilized to pull 997,046 water temperature recordings. The data was then aggregated into hourly means to form a consistent time series.

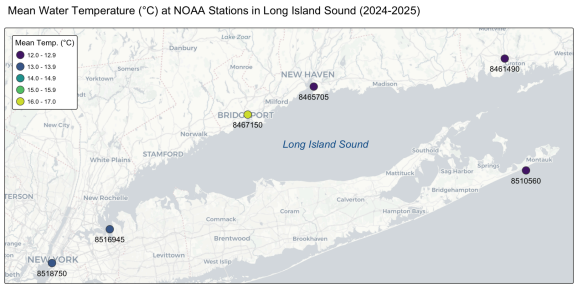

Fig. 1 Map of Long Island Sound with the six selected NOAA stations and their mean water temperatures over the study period.

Water temperature in coastal systems such as Long Island Sound exhibits both temporal dependence, which is a change over time, and spatial dependence, which is similarity between nearby stations. The geographic distribution of the stations, as seen in Fig. 1, prompts the question of whether their distance from one another influences temperatures or if they are far enough apart to be able to accurately measure and forecast future temperatures with data isolated from a single station. Thus, the analysis of data from one primary station determines a baseline for comparison using an ARIMA model. An STARIMA model, which takes into account the spatial relation with neighboring stations, is subsequently applied.

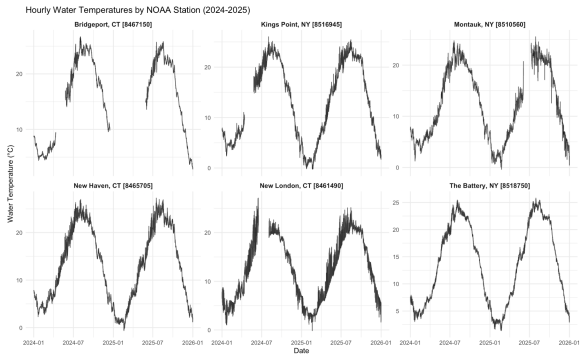

Fig. 2 Hourly mean water temperature for each of the six NOAA stations is plotted.

An initial review of the hourly temporal data, as seen in Fig. 2, revealed significant gaps in recorded temperatures at the Bridgeport, CT, station [8467150]. To maintain the integrity of the spatio-temporal matrix, this station was removed from subsequent analysis and modeling, leaving five active stations. The New Haven, CT, station [8465705] was selected as the primary reference station for exploratory data analysis and baseline modeling due to its central location in the Sound and high data completeness.

2. Exploratory Spatio-Temporal Data Analysis

Traditional time series models such as ARIMA are effective at capturing temporal patterns while assuming that each location is independent. The initial EDA ignored spatial dependencies in order to focus on the selected station. To also satisfy the continuity requirements of ARIMA, which rely on evenly spaced, uninterrupted temporal matrices,

missing values in the data required linear interpolation (Box et al., 2015). Linear interpolation is a commonly used and computationally efficient method for handling small gaps in most time series, including environmental data (Moritz & Bartz-Beielstein, 2017). The New Haven station was missing temperature values for 33 hours, representing less than 0.2% of the total temporal record. Linear interpolation (na.approx in R) was then used to fill missing values for all stations.

To understand the underlying patterns driving the water temperature, a Seasonal, Trend, and Loess (STL) decomposition was applied to a two-week summer sample from the New Haven station (Fig. 3). A macro-scale plot of the high-frequency data would obscure the repeating diurnal patterns. Following standard exploratory practices for hourly time series, a subseries spanning two weeks was analyzed in order to avoid excessive noise (Hyndman & Athanasopoulos, 2018). This revealed a deterministic 24-hour seasonal cycle driven by solar heating, in which temperatures rise during the day and cool at night. The daily cycle occurs alongside a macro-level temperature trend.

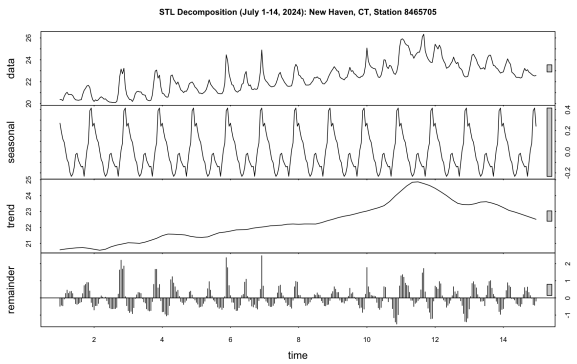

Fig. 3 STL decomposition for the New Haven, CT, station during two weeks in July 2024.

The presence of seasonality led to plotting the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the raw data to confirm non-stationarity and get a sense of possible ARIMA values for (p, d, q) and (P, D, Q), non-seasonal and seasonal parameters, respectively. The ACF of the raw data in Fig. 4 exhibits the slow decay that is characteristic of non-stationary data. The PACF of the raw data confirms a strong Autoregressive (AR) signature, with possible AR(1) or AR(2) values because of the first two spikes. One can expect the best predictor of the water temperature right now to be the water temperature one or two hours ago, which is indicative of water's thermal inertia.

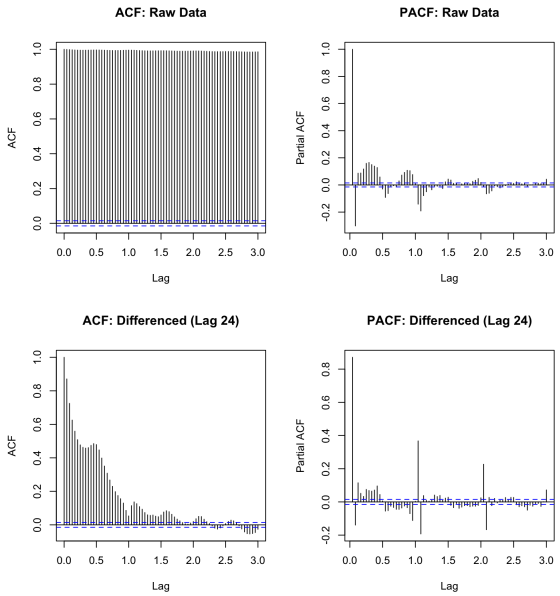

Fig. 4 Autocorrelation and Partial Autocorrelation Functions for raw data (top row) and differenced data with a lag of 24 hours (bottom row).

Applying a 24-hour seasonal difference removed the primary daily cycle, as seen in the two lower plots of Fig. 4, yielding a much more stationary series appropriate for ARIMA modeling. This is clear in the ACF of differenced data as it decays downward. The PACF of differenced data still has spikes at fixed seasonal lags, which can indicate a seasonal moving average (SMA) is likely needed in the ARIMA parameters. A standard univariate ARIMA model was fitted to the New Haven Station data using the auto.arima function in R.

3. Methodology & Results

After exploring the data, an ARIMA model was fitted to the data to construct a baseline model of the New Haven station in isolation from its neighbors. Understanding the interconnected nature of the Sound and environmental data, an STARIMA model can then be fitted and compared. An out-of-sample 14-day window covering January 1-14, 2026, was selected to test both models. Plotting each model and calculating their Root Mean Square Error (RMSE) will show which model is a better predictor of future water temperatures.

The seasonal ARIMA model is typically written as *SARIMA(p, d, q)(P, D, Q)S*. The (p, d, q) parameters are non-seasonal, (P, D, Q) are seasonal, and S is the lag of the seasonal component. The auto.arima function in R produced the model ARIMA(5,1,2)(2,1,0)[24]**.** Immediately, one notices the previous assumption made of an SMA value (Q) being needed was incorrect. Instead, the values can be interpreted in the following manner:

Non-seasonal parameters

p = 5 Autoregression looks back 5 hours to predict current temperature.

d = 1 Differences by 1 lag (of 1 hour).

q = 2 Looks back 2 hours at forecast errors to adjust the prediction.

Seasonal parameters

S = 24 Seasonal frequency is every 24 hours, meaning each seasonal cycle is one day.

P = 2 Autoregression looks back 2 days to predict current temperature.

D = 1 Differences by 1 lag (of 1 day).

Q = 0 No seasonal moving average.

The model produced an RMSE of 0.246125 °C, used to compare to that of the RMSE produced when the model is tested. In Fig. 5, the ACF of the residuals is relatively clean with some minor seasonal echoes in the PACF.

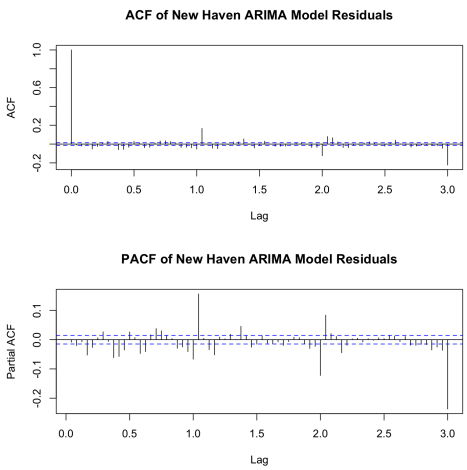

Fig. 5 ACF & PACF of the residuals from ARIMA(5,1,2)(2,1,0)[24] for the New Haven data.

The Ljung-Box test resulted in a low p-value of 2.2e-16, which is considered statistically significant ($p < 0.05$). It's important to note the 'p-value problem' associated with very large samples ($n > 10{,}000$), where p-values quickly go to zero and lose their practical significance (Lin et al., 2013).

Testing the model on the 14-day window in January 2026 produced the plot in Fig. 6 and an RMSE of 3.978 °C. This number is higher than the training data RMSE, suggesting overfitting. The model may be too rigid and not able to capture or learn certain trends. While the model forecasted the general structure of the temperature curve at the start, it struggled to capture the upward trend after a couple of days.

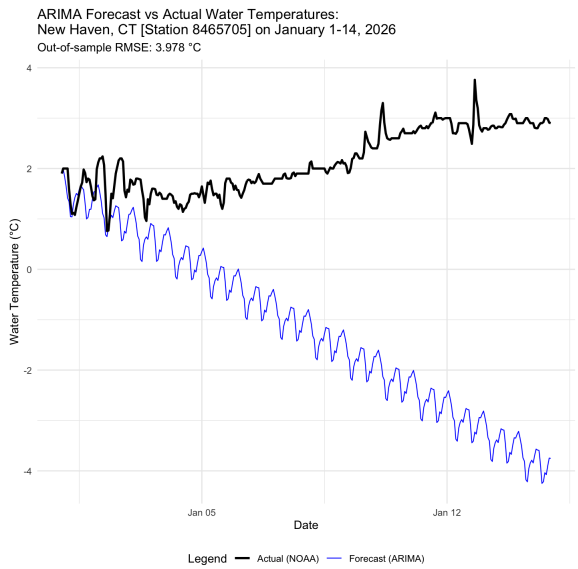

Fig. 6 ARIMA forecast tested against actual water temperatures recorded at the New Haven Station in early January 2026.

Spatial Dependence

Remembering Fig. 1, it's easy to see that the NOAA stations are close enough and connected by a large body of water to consider their spatial relations. Especially when dependencies in the Sound are driven by atmospheric conditions, tidal mixing, and coastal circulation processes (O'Donnell et al., 2014). The univariate approach failed to account for spatial spillover effects moving through Long Island Sound. Fluctuations at one station inevitably influence neighboring stations with a time delay. Since the ARIMA model cannot capture these interactions, a Space-Time ARIMA (STARIMA) model was developed.

In order to account for spatial dependence, a spatial weights matrix (W) was needed. An Inverse Distance Weighting (IDW) approach was chosen over an adjacency matrix due to the nature of the observational data. Distance-based spatial weighting accurately reflects the continuous fluid dynamics and thermodynamics of a marine environment since the stations are points in water and not large polygons with shared boundaries. The IDW method operates on the principle of distance decay and Tobler's First Law (1970), asserting that the oceanographic and meteorological conditions at one station will exert a stronger influence on a nearby station than on a distant one. To construct the matrix, the geographic coordinates of the NOAA stations were first projected into the best suited coordinate reference system (CRS) for Long Island Sound, which is EPSG:2263 (NAD83 / New York Long Island). Pairwise distances were calculated in US Survey Feet. Ultimately, the unit does not affect the final spatial weights matrix.

Matrix W was calculated as the inverse of these distances which were row-standardized. A quick visual check with the map in Fig. 1 shows the two stations closest to each other are at the western end of the Sound at Kings Point, NY [8516945] and The Battery, NY [8518750]. Fig. 7 shows these two stations having the shortest distance between them at under 100,000 feet.

```
--- Pairwise Distance Matrix (US Survey Feet) ---
        8518750 8516945 8465705 8461490 8510560
8518750       0   78770  370139  581708  566685
8516945   78770       0  291601  503362  492774
8465705  370139  291601       0  224928  266728
8461490  581708  503362  224928       0  133491
8510560  566685  492774  266728  133491       0
```

Fig. 7 Output from R of matrix of pairwise distances among the five active stations. The first row and column are the station IDs.

Fig. 8 shows a spatial weight above 0.6 between the two closest stations, which is the highest weighting in the matrix. The accurate calculation of W allows the STARIMA model to account for external influence from nearby locations. Furthermore, because the IDW method does not define higher-order neighbors, the spatial lag term remains 1. There are no neighbors of neighbors to worry about, like with an adjacency matrix. The influence of each neighbor scales strictly by its geographic proximity.

```
--- Row-Standardized Spatial Weight Matrix (W) ---
        8518750 8516945 8465705 8461490 8510560
8518750   0.000   0.672   0.143   0.091   0.093
8516945   0.630   0.000   0.170   0.099   0.101
8465705   0.189   0.239   0.000   0.310   0.262
8461490   0.110   0.127   0.284   0.000   0.479
8510560   0.117   0.135   0.249   0.498   0.000
```

Fig. 8 Spatial Weight Matrix (W) output from R. The first row and column are the station IDs.

Utilizing W, the Spatio-Temporal ACF and PACF (ST-ACF, ST-PACF) could be plotted from the seasonally-differenced space-time matrix to get a sense of the STARIMA parameters. To determine the parameters for *STARIMA(p, d, q)*, the plots in Fig. 9 were visually inspected. ST-PACF revealed a significant spike at the first lag followed by a drop into the confidence interval, indicating an autoregressive order of p = 1. ST-ACF showed a slow decay toward 0, indicative of an autoregressive process and a moving average order of q = 0. Based on these observations, STARIMA(1, 24, 0) was identified as the best starting structure for fitting a model with 24-hour seasonal difference to the training data.

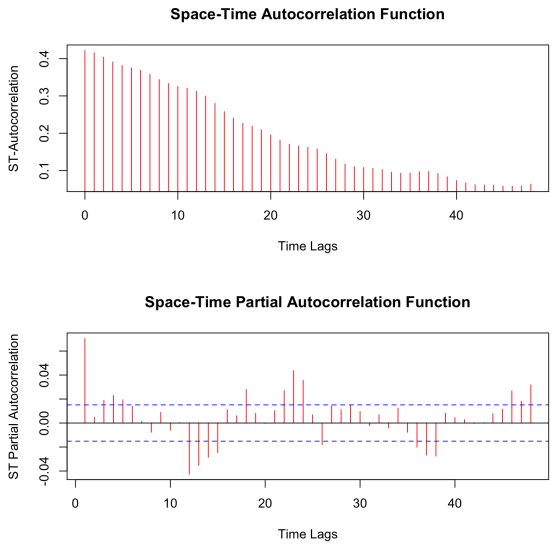

Fig. 9 ST-ACF & ST-PACF of the seasonally-differenced space-time matrix made from data from the five active stations around Long Island Sound.

While STARIMA(1, 24, 0) produced an RMSE of 0.182 °C, already lower than that of ARIMA(5,1,2)(2,1,0)[24], empirical tuning and out-of-sample performance checks revealed that increasing p = 1 to p = 2 yielded a lower RMSE on the test data. STARIMA(2, 24, 0) is a better fit that resulted in an RMSE of 0.145 °C. To evaluate the adequacy of the final STARIMA model, the study used a Ljung-Box Test and plotted an ST-ACF of the residuals. The Ljung-Box Test in R again resulted in p-value < 2.2e-16. With a sufficiently large sample, any minuscule correlation will become statistically significant, so the magnitude of the effect should be looked at rather than the confidence bounds (Sullivan & Feinn, 2012). While Fig. 10 shows minor remaining correlations breaking the bounds of the confidence interval, the absolute values of these correlations were trivial at less than 0.05. While it's possible other factors beyond the spatial-temporal produce these correlations, the residuals approximate white noise for the purpose of this study. As such, STARIMA(2, 24, 0) remained the final model and its predictive accuracy was tested against actual water temperatures in January 2026.

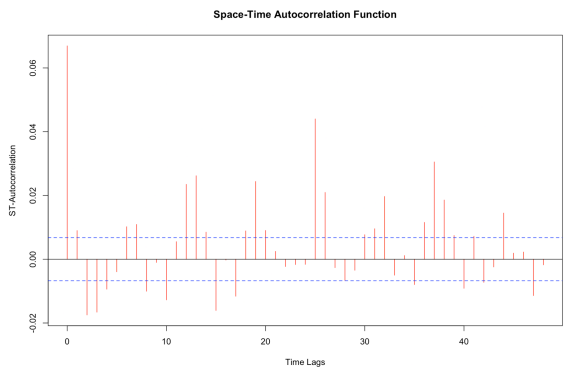

Fig. 10 ST-ACF of the residuals of the chosen model, STARIMA(2,24,0).

The STARIMA model was tested on the same 14-day window in January 2026 that the ARIMA model was tested on. The STARIMA forecast drastically outperformed the baseline ARIMA model. Fig. 11 shows how the STARIMA forecast for water temperatures at the New Haven station captured the true amplitude of the temperature swings much more accurately than the baseline model.

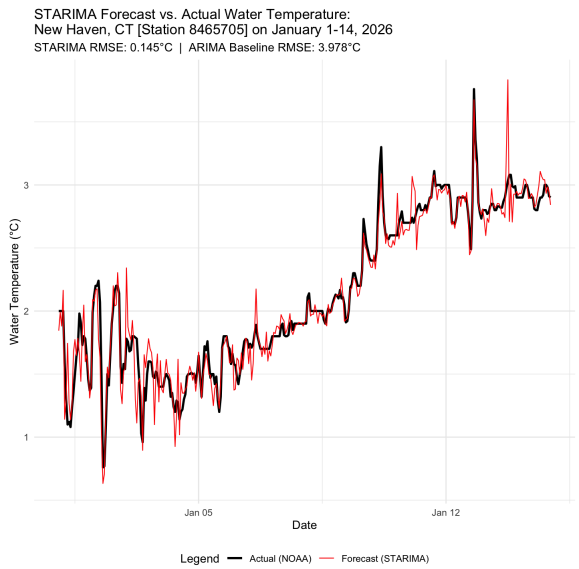

Fig. 11 STARIMA forecast tested against actual water temperatures recorded at the New Haven Station in early January 2026.

Furthermore, the model proved robust across the entire network. Fig. 12 shows the STARIMA model's out-of-sample forecasting performance for the remaining four active stations on the same dates of January 1-14, 2026. The graphs show close alignment with the actual NOAA observations across the four locations.

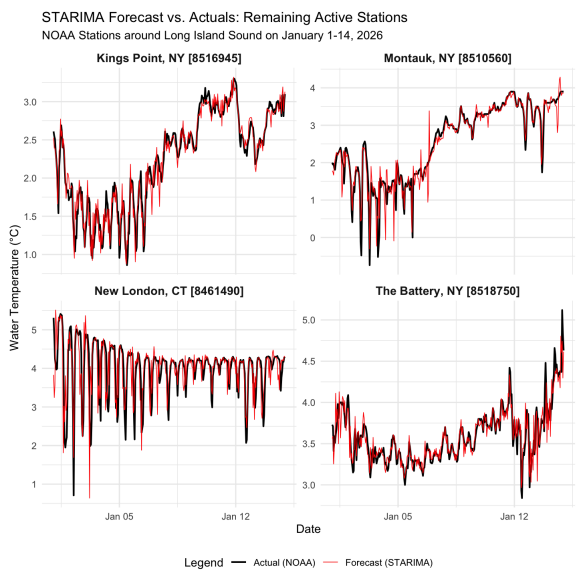

Fig. 12 STARIMA forecasts tested against actual water temperatures recorded in early January 2026 at the other four active stations.

4. Discussion & Conclusion

This study compared the performance of an ARIMA model and an STARIMA model in forecasting water temperatures around Long Island Sound. The results clearly demonstrate that incorporating the spatial dimension substantially improves predictive performance in this environmental setting.

When comparing a baseline ARIMA model to the STARIMA model, both model fit and forecasting accuracy improved. The RMSE of the ARIMA(5,1,2)(2,1,0)[24] model when applied to the test window was 3.978 °C, which is much higher than the RMSE of the STARIMA(2,24,0) at 0.145 °C over the first 14 days of 2026. A 96% reduction in the RMSE. While ARIMA generally can be effective at modeling time series data, it struggles to generalize in environments where external spatial influences play a major role. Meanwhile, the selected STARIMA model was able to closely track observed temperature dynamics across all active stations. Such a dramatic improvement from taking into account spatial spillover effects was possible through the incorporation of a spatial weights matrix. Simply stated, the STARIMA model captured dynamics that a purely temporal model could not.

While the STARIMA model provided greater predictive accuracy for a two-week test window, the strengths of the ARIMA model are more related to the process and methods of the research study. It offers simplicity, interpretability, and computational efficiency. Implementation was straightforward using auto.arima in R and it could be a good enough model if only forecasting very soon into the future on the scale of minutes or hours. In contrast, STARIMA was more complex to implement, requiring a spatial weights matrix and additional diagnostic tools like ST-ACF and ST-PACF. In the end, for this study, the complexity was justified by the substantial gains in forecast accuracy.

Model performance was also consistent across the active stations around Long Island Sound. The STARIMA model improved forecasts at the primary station in New Haven and produced accurate forecasts across the remaining active stations. This indicates that the spatial weights captured by the IDW approach are appropriate for the purposes of representing interactions within the Sound.

Despite the strong performance of the final model, several limitations can be acknowledged. First is the small number of stations used. Five stations, after removing the Bridgeport station, limits the amount of spatial information available to the model. The active stations may not fully capture the complexity of the marine environment in the region. The possibility of including more NOAA stations exists, but this spreads across a larger and less contained geographic area. Pulling more time data for each station is also possible, but this can introduce more missing values. This leads to the second limitation, the use of linear interpolation to fill gaps in recorded values. Even though minimal interpolation was used in the dataset, smoothing and reduced variability could have been introduced. Third, the spatial relationship between each station was oversimplified. The spatial weights matrix was assumed to be static and based on linear distances, when in reality coasts, currents, and waterways are dynamic and non-linear. The complexity of the real world introduces changing environmental and human-made conditions.

Applying more sophisticated methods in R would improve future work for similar studies. Kalman smoothing or spatio-temporal interpolation could be tried for filling missing values. The incorporation of more variables, such as air temperature, pressure, current speed, and other known environmental factors or pollutants that are present in the area of study, whether the Sound or elsewhere, could help account for further variability in water temperatures.

In conclusion, this study highlights the importance of accounting for spatial dependence in environmental time series analysis. The STARIMA model, by integrating both space and time, offers a more accurate and realistic framework for forecasting water temperatures in Long Island Sound. Spatio-temporal models help predict environmental processes, like water temperature changes, which have real-world implications on fishing and the livelihoods of coastal communities.

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